

Zachary Virgilio

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Summary

Mathematics PhD combining deep expertise in statistical modeling, time-series analysis, and machine learning with hands-on experience building production-grade RAG systems, transformer architectures, and agentic AI pipelines. Proven ability to translate rigorous quantitative foundations into scalable, real-world AI solutions for forecasting, anomaly detection, and data-intensive decision-making.

Technical Skills

- **AI & Machine Learning:** PyTorch, Transformers, LLM Foundations, RAG (Retrieval-Augmented Generation), Agentic Systems, Time-Series Forecasting, Anomaly Detection, Geometric Deep Learning
- **Data Science & Engineering:** Python (Pandas, NumPy), Statistical Modeling, Data Visualization, Monte Carlo Simulation, Scikit-learn
- **Scientific Computing & Mathematics:** Stochastic Processes, Optimization, Linear Algebra, High-Dimensional Analysis, Power Analysis, Sensitivity Analysis
- **Software Engineering & Tools:** Python, C++, R, Git, FastAPI, ChromaDB, Sentence Transformers, Hugging Face, AI-Assisted Development (Claude Code)

Machine Learning Projects

AI-Powered Biotech Research Assistant (RAG & Agentic System)

In Progress

- Built a production RAG pipeline using ChromaDB for vector storage, Sentence Transformers for embeddings, and cosine similarity retrieval across internal research documents and clinical trial PDFs.
- Orchestrated multiple agents to query the web, ClinicalTrials.gov, and SEC EDGAR; exposed tools as MCP servers integrated with an Anthropic API-powered application.

Neural Memory & Test-Time Learning (Titans Implementation)

2026

- Implemented the Neural Memory module from the *Titans* paper, enabling recurrent state updates for long-horizon reasoning tasks.
- Engineered stateless modules using functional transformations (PyTorch `vmap`, `grad`) to optimize recall and gradient flow.

GPT-2 Foundation Model Implementation

2025

- Built GPT-2 from scratch in Python and PyTorch, manually implementing Multi-Head Attention and Flash Attention for training throughput optimization.
- Applied numerical stability and optimization techniques: loss scaling, weight tying, and advanced sampling strategies.

Research Experience

Doctoral Researcher, UC Riverside

2020–2025

- Specialized in **Differential Geometry** and **Lie Theory**; applied high-dimensional analysis and optimization to study stability of complex geometric structures.
- *Key Achievement:* Developed the first standardized procedure for determining Anosov representations from $SU(n+1, 1)$ - and $SO(2, n+2)$ -Higgs bundles, replacing ad-hoc case-by-case analysis.

Research Assistant, High-Frequency Trading, Washington University in St. Louis

2016–2017

- Developed Python tools to parse and extract time-series statistics from terabyte-scale NASDAQ ITCH datasets.
- Engineered robust, low-latency data ingestion pipelines for tick-level data, optimizing throughput for real-time anomaly detection and time-series analysis.

- Statistical Analysis Intern, Partner Funds Management LP** 2015
- Built a large-scale Monte Carlo simulation in R modeling a pharmaceutical clinical trial from interim data, varying effect-size assumptions to perform sensitivity and power analysis.
 - Results increased fund confidence in trial success; the fund grew its position and the trial succeeded, generating significant returns.

Education

- PhD in Mathematics, UC Riverside** 2025
- GPA 3.98/4.00 Thesis: *Geometric Structures Arising from $SU(n+1,1)$ -Higgs Bundles*
- MA in Mathematics, University of Denver** 2020
- GPA 4.00/4.00
- BA in Mathematics, Minor in Computer Science, Washington University in St. Louis** 2018
- Graduated with Highest Distinction